13 August 2026

The Editors
Journal of Computer Languages
Elsevier

Dear Editors,

I submit for your consideration my manuscript, **"Grammar-Derived Operation Abstraction for Cross-Project Vulnerability Detection"**, for publication as a research article in the *Journal of Computer Languages*.

What the paper is about. The paper studies a program abstraction: a labelling function over the node alphabet of a C grammar, and the question of what a node in a program graph should carry when a learned analysis is expected to transfer to code it has never seen. I compare, under strict experimental control, three node representations — project-specific lexical tokens, an engineered taxonomy of operation types constructed from API allow-lists and identifier rules, and the parser's own grammar node kinds.

Why I believe it fits this journal. The object of study is a language-level artefact rather than a machine-learning architecture. I specify the abstraction as a total function over a tree-sitter grammar alphabet, give the projection lattice relating its granularities with a nesting proof, and — unusually for this application area — evaluate it *as a static analysis*, reporting per-class precision and recall against annotated ground truth. That evaluation is not decoration: it uncovered three defects invisible to downstream accuracy, including a rule that classified pointer writes as deallocations and a class that counted 655 stream-I/O call sites as memory operations.

The principal finding. The representation matters more than the architecture, and the representation that works is the one that costs nothing. Grammar node kinds improve cross-project ranking by +0.078 AUC ($p = 0.007$) and +0.071 AUPRC ($p = 0.002$) over lexical tokens, winning every project-level fold, whereas the three message-passing schemes I compare differ by less than half as much under a fixed representation. The engineered taxonomy — the artefact I built first and expected to be the contribution — is statistically indistinguishable from lexical features ($p \approx 0.9$). I report this negative result at the same length as the positive one, and explain it: an information-theoretic analysis shows the taxonomy is 98.2% predictable from the grammar kind alone ($H(\phi \mid \kappa) = 0.06$ bits), because most of its rules are syntax-directed and therefore already decided by the parser. I recommend that redundancy check as routine practice before an abstraction is credited with an effect.

Methodological care. Cross-project evaluation in this area is easy to get wrong, so I control for the failure modes documented in the literature: near-duplicate functions are removed before splitting; projects are the unit of partition, with repositories too large to place without dominating a fold held train-only and reported; vocabularies are fitted on training projects only; and confidence intervals come from a cluster bootstrap that resamples held-out projects rather than functions. Every table reports the trivial all-positive classifier as a floor.

Originality and availability. The manuscript is original, is not under consideration elsewhere, and has not been published previously. All code, annotations, and the scripts that regenerate every reported number are publicly available.

I confirm that the manuscript has been approved for submission, that it is not under consideration elsewhere, and that I have no competing interests to declare.

Thank you for your consideration.

Yours sincerely,

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam
Email: `vinh.dq4@buv.edu.vn`
ORCID: 0000-0002-3877-8024